Beyond Forecast Accuracy: Comparing Probabilistic Demand Forecasts by the Inventory They Require across Retail Settings and Periods

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ABSTRACT

The forecast with the lowest error is not necessarily the forecast that supports the lowest inventory. We test this using seven probabilistic methods—three statistical approaches and four gradient-boosting variants—in a weekly order-up-to simulation with optional liquidation, on all eligible series from M5 (physical stores) and VN1 (multi-vendor e-commerce) over three 26-week windows per dataset. Because inventory costs are unobserved, we compare stock requirements at equal realised fill rates with bootstrap confidence intervals. On M5, LightGBM quantile regression (LGB-Q) has the lowest volume-weighted quantile loss and, at a fill rate of 0.94, needs 3–11% less inventory than the safety-stock, conformal, TSB-NB and exponential-smoothing alternatives in each window; this holds across the tested lead times and LightGBM settings. VN1 ranks the methods differently in different windows: TSB-NB needs 13–16% less inventory for stable-demand series but fails to reach a fill rate of 0.92 on growing-demand series in two windows, and neither ADI–CV² classes nor clusters consistently identify the right method in advance. A median of all seven methods' quantiles, specified before testing, needs 7–8% less VN1 inventory than LGB-Q at a fill rate of 0.92, but the saving loses significance when warehouses are resampled, does not persist when τ is calibrated from past weeks only, and is not confirmed on an independent footwear dataset, where LGB-Q again needs the least stock. The N-HiTS baseline leads no inventory comparison. Quantile-based liquidation changes fill rate by at most 0.02 percentage points but shows no consistent economic advantage over 26 weeks under the assumed costs.

CCS CONCEPTS • Applied computing → Forecasting; Supply chain management • Computing methodologies → Machine learning; Cluster analysis

Additional Keywords and Phrases: probabilistic forecasting; inventory management; intermittent demand; quantile regression; demand segmentation; liquidation; M5; VN1

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1 INTRODUCTION

An order based on the expected value of demand cannot describe the risk of a stockout. That risk matters: insufficient inventory loses sales, while excess stock incurs holding costs and may later be liquidated. Forecast quantiles allow an order-up-to level to be set for a chosen service target [3]. Whether the quantiles are statistically accurate, however, is a separate question from how much stock their use requires.

The difference matters when selecting a forecasting method. M5 ranks forecasts using weighted RMSSE and weighted scaled pinball loss [1]; its uncertainty track was not designed around an inventory decision [3]. Work with VN1 has likewise focused on accuracy and computation [4]. Inventory simulations can yield a different ranking [5, 6, 23]. In particular, Theodorou et al. [7] report discrepancies on M5 when lost sales are costly, especially for intermittent series. They identify online retail as a setting in need of further testing.

Three features of the existing evidence motivate our design. First, M5 inventory evaluations have used selected subsets: 8,000 high-volume series [8], volatile food products [9], or series with at least 2.5 years of history [7]. These restrictions leave some sparse series outside the analysis. Second, most comparisons use a single dataset and test period [5, 7–9, 23], so period-to-period variation is poorly characterised. Third, public sales panels do not provide historical inventory positions or unit costs. Any simulated monetary outcome must consequently depend on assumptions about those missing quantities [7].

We hold the inventory policy fixed and vary only its demand forecasts. Seven methods are evaluated on every eligible series in M5 and VN1, with weekly decisions simulated over three 26-week windows per dataset and inventory compared at matched realised fill rates. Our contributions are as follows.

- Ranking instability across settings and periods. Accuracy and inventory rankings agree on M5, where LGB-Q needs 3–11% less stock than the point-forecast, conformal, TSB-NB and exponential-smoothing alternatives in every window, but not on VN1, where the leading method changes between windows.

- Limits of prospective method selection. Pre-test demand classes, clusters and the previous window's leader do not reliably identify the best method, and reweighting VN1 to M5's feature distribution leaves the VN1 findings unchanged.
- Retrospective efficiency versus prospective reliability. Savings measured at matched realised fill need not survive deployment-style evaluation: with service calibrated from past weeks only, warehouse-level resampling and an independent footwear panel, the 7–8% VN1 saving of a pre-specified seven-method median disappears, whereas the M5 ranking holds. With τ chosen on the previous window, the direct quantile learners stay closest to the target fill.
- Decision guidance and robustness. A break-even salvage ratio states when liquidation pays, and checks on lead time, hyperparameters, censored demand and computing cost show which findings are stable. Code and results are public.

2 RELATED WORK

Retail forecasting benchmarks. In M5's accuracy track, all 50 highest-ranked teams used LightGBM, and the winning entry combined models fitted with Tweedie loss [2]. The leading uncertainty-track entry trained a separate LightGBM model for each quantile [3]. We therefore include both point-based and directly estimated quantiles. Zanotti [4] compared global-model ensembles on M5 and VN1 in terms of forecast accuracy and computing cost, without simulating stock decisions.

Intermittent demand. TSB updates the probability of non-zero demand from one period to the next, allowing the forecast to decline when sales stop [10]. Average inter-demand interval (ADI) and squared coefficient of variation of positive demand sizes (CV^2) are conventional descriptions of demand intermittency [11]. For M5 we use class boundaries ADI = 4/3 and $CV^2 = 0.5$ [1]. Damato et al. [12] found no established global-model architecture that dominates on intermittent data.

Forecasts used for inventory. Goltsos et al. [6] document the frequent separation of forecasting studies from the replenishment decisions they inform. Kourentzes et al. [13] fit exponential smoothing with an inventory-based criterion; van der Haar et al. [14] estimate order quantities directly. On M3 series, Petropoulos et al. [24] found forecast combinations effective in a rolling order-up-to simulation. Evidence from M5 is mixed: simple empirical quantiles perform well in the 28-day daily-review experiment of Spiliotis et al. [23]; Wang et al. [5] study forecast combinations with a single-period newsvendor; Zabraoui et al. [9] apply reinforcement-learning stock policies to selected products; and Mohammed et al. [8] evaluate safety stock for high-volume series with an illustrative cost calculation.

Theodorou et al. [7] are the closest point of comparison. They simulate lost-sales inventory control for one year on 7,783 daily M5 product–state series. Their design covers 13 forecasting methods and 24 service-target, review-period and lead-time combinations. Quantiles are constructed from point forecasts and past forecast errors. Although error and inventory trade-off curves generally align, inventory-cost rankings can favour less accurate methods when stockouts carry a high penalty.

We retain eligible series across demand classes, use two retail settings with three test windows plus an independent third panel, and include a liquidation decision. We also estimate quantiles directly rather than deriving all of them from residuals. Inventory is compared at the same realised fill rate, because a common nominal quantile level does not give all methods the same achieved service [7]. Neither multi-period stock simulation nor decision-based forecast evaluation is new [5, 7, 9, 14]; our contributions are the comparison across datasets, periods and decision settings and a check of whether retrospective inventory savings survive deployment-style evaluation.

3 METHOD

3.1 Data

M5 [1] reports daily unit sales for 3,049 products in 10 Walmart stores, giving 30,490 product–store series. Aggregating complete weeks yields 277 weekly observations. VN1 [21] contains 15,053 client–warehouse–product series spanning 196 weeks of multi-vendor e-commerce sales; despite its name, the dataset is not from Vietnam. Eligibility is determined separately for each test window: a series must start at least 13 weeks before that window and record at least one earlier sale.

Table 1. Dataset characteristics measured before the main test window.

Property	M5	VN1
Retail setting	One retailer; 10 physical stores	Multi-vendor e-commerce; 46 vendors; 328 warehouses
Total series	30,490	15,053
Eligible series (main / second / third)	30,381 / 29,917 / 28,824	13,844 / 11,442 / 9,383

Property	M5	VN1
Number of weeks	277	196
Zero-demand weeks since series start	24.4%	49.1%
Smooth / erratic / intermittent / lumpy	53.0 / 5.8 / 33.9 / 7.3%	18.8 / 13.7 / 47.2 / 20.3%
Additional variables	Prices, events, SNAP days, product hierarchy	Prices in weeks with sales only

A single processing pipeline handles both panels. The source data differ in the definition of a series start, but the resulting dates agree for 99.5% of M5 series. Their demand characteristics are nonetheless distinct (Table 1): a classifier using nine demand features identifies the dataset with an AUC of 0.91 (repository); Section 5.7 tests whether such differences explain the results.

Independent panel. To check the combinations on data not used to specify them, we add the retail-channel sales of a Vietnamese footwear chain, public on Kaggle (tienanh2003/sales-and-inventory-snapshot-data): 1,001 style–colour series over 82 weeks (January 2022 to July 2023), excluding returns and three partial weeks. With the eligibility rule above, 909 series enter one 26-week test window from 30 January 2023. Forecasts use the same pipeline, 21 features, refit schedule and policy; rerunning it on the processed sales file reproduces them exactly. Only one window fits the short history, month-end stock snapshots are unused, and single-method results on this panel existed before, but were not used for, specifying the combinations.

3.2 Forecast targets and methods

Each week, we predict quantiles $q \in \{0.50, 0.80, 0.90, 0.95, 0.99\}$ of demand accumulated over horizon h. Replenishment uses h = 3 weeks and liquidation h = 13 weeks. Quantiles are fitted to cumulative demand rather than obtained by summing separate weekly quantiles, which would generally be incorrect.

EMP: empirical quantiles of rolling h-week demand totals observed during the preceding 104 weeks.

ETS: simple exponential smoothing and a normal approximation for the h-week demand total [18].

TSB-NB: TSB [10] with a negative-binomial predictive distribution [25], matched to the estimated mean and variance of h-week demand.

LGB-T: a global LightGBM point model fitted with Tweedie loss [15], with a normal safety-stock adjustment derived from validation residuals for each series.

LGB-C: the same point model, with split-conformal quantiles [17] based on residuals pooled within demand classes.

HGB-Q: histogram gradient boosting [19] fitted with quantile loss to 300,000 sampled training rows. It provides a second direct-quantile implementation.

LGB-Q: LightGBM quantile regression [16], fitting a separate global model for each quantile.

An additional combination, C2, specified post hoc after inspecting the results, averages the corresponding LGB-Q and TSB-NB quantiles (Section 5.5). That section also evaluates two combinations of all seven methods fixed before they were run: the mean (E7) and the median (E7-med) of their quantiles. All four global gradient-boosting methods have the same input categories: sales lags, rolling mean and variation, proportion of zero-sales weeks, time since last sale, calendar terms, price and series scale. Targets are scaled to the series level. M5 additionally supplies events, SNAP days and product hierarchy. No hyperparameter search is performed. LGB-Q, LGB-T and LGB-C share one LightGBM configuration: learning rate 0.05, 63 leaves with no depth limit, at least 200 samples per leaf, L2 penalty 1, feature and row subsampling fractions of 0.8, and at most 1,000 boosting rounds with early stopping after 50 rounds. Each refit trains on up to 104 weekly origins and validates on the following 13; LGB-T and LGB-C use Tweedie loss with variance power 1.1. HGB-Q uses learning rate 0.1, at most 300 iterations, 63 leaves and at least 200 samples per leaf. Sensitivity checks appear in Section 5.8.

3.3 Inventory decision policy

We simulate weekly review with a separate lost-sales inventory process for each series. At the start of a week, orders from L weeks earlier arrive. Inventory position IP includes on-hand and outstanding units. If liquidation applies, stock above threshold T is removed. The replenishment order is $\max(S - IP, 0)$, where S is the τ-quantile of demand over L + 1 weeks; no new order is placed in a week with liquidation. Demand then occurs, and any unmet units are recorded as lost sales. Unless otherwise stated, L = 2 weeks.

Four policies share the replenishment forecasts: no liquidation; quantile liquidation, with T set to the method's 0.95-quantile of 13-week demand; a fixed threshold of 26 times mean weekly sales in the preceding 26 weeks; and dead13, which removes all stock after 13 consecutive weeks without sales and suspends ordering until a new sale is recorded.

3.4 Evaluation metrics

Forecast accuracy. For a series, scaled quantile loss divides pinball loss by the in-sample naïve forecast error. A small denominator makes its cross-series mean unstable on VN1 (Section 5.1). We therefore report volume-weighted quantile loss (wQL: total pinball loss divided by total demand), the median series-level scaled loss, and the mean rank of each method across series. Rank differences are evaluated with the Friedman test and Nemenyi critical difference [20].

Inventory. Each 26-week window starts with a four-week warm-up, leaving 22 weeks for evaluation. Fill rate is the fraction of demanded units sold; inventory is measured in weeks of demand. We vary the order-up-to quantile level τ over the five reported values, then interpolate along each inventory–fill-rate curve to find the stock required at a specified realised fill rate [7, 25]. A method's frontier rank is its mean inventory rank across fill targets of 0.90–0.98. The 200 series-level bootstrap resamples reflect uncertainty across series, not uncertainty about performance in future periods.

Liquidation. For each policy we simulate each series twice from the same starting stock, with and without liquidation, and write Δ for the difference between the two runs over the test window. With unit cost normalised to 1, ΔO is the change in units ordered, ΔS the change in units sold, ΔI the change in inventory (unit-weeks), ΔE the change in the end-of-horizon position (on-hand plus on-order units, valued at cost) and Q the number of units liquidated. With weekly holding rate h (annual rate / 52) and markup m (selling price $(1 + m)$ times cost), liquidating at salvage ratio s is at least as profitable as holding when $s \cdot Q + (1 + m) \cdot \Delta S - \Delta O - h \cdot \Delta I + \Delta E \geq 0$. The break-even salvage ratio is therefore $s^* = [\Delta O - \Delta E + h \cdot \Delta I - (1 + m) \cdot \Delta S] / Q$. A value $s^* < 1$ means that liquidating below cost still pays off; $s^* > 1$ means that liquidation pays off only if stock can be sold above cost. Valuing the end position at cost favours holding, so the reported s^* is an upper bound. Calculations cover annual holding costs of 10–40% and markups of 30–100%. Neither actual salvage prices nor inventory costs are observed in these datasets.

4 EXPERIMENTAL SETUP

For M5 and VN1, the main window comprises the final 26 weeks; two earlier windows are formed by removing the last 26 and 52 weeks, respectively. The footwear panel has only its final 26-week window. Within each window, the models are refitted at the beginning of two 13-week blocks, while forecasts and inventory decisions are generated weekly. Training is repeated for every window. Experiments ran on an Intel Core i5-12450H laptop CPU with 16 GB of RAM. Code and result files: https://github.com/anhduyBA/forecast-to-decision-benchmark.

5 RESULTS

5.1 Forecast accuracy

Table 2. Three-week forecast accuracy, by dataset and window. Lower values indicate better performance; wQL is volume-weighted quantile loss.

Method	Window	M5 wQL	M5 median	M5 rank	VN1 wQL	VN1 median	VN1 rank
EMP	main	0.115	0.174	4.85	0.172	0.202	4.72
EMP	second	0.120	0.176	4.88	0.146	0.207	4.91
EMP	third	0.126	0.176	4.95	0.197	0.239	4.90
TSB-NB	main	0.088	0.155	4.22	0.129	0.152	3.25
TSB-NB	second	0.091	0.150	3.99	0.088	0.150	3.30
TSB-NB	third	0.098	0.150	4.12	0.137	0.186	3.31
ETS	main	0.084	0.153	4.12	0.129	0.174	4.50
ETS	second	0.089	0.152	4.09	0.092	0.173	4.54
ETS	third	0.095	0.150	4.12	0.136	0.198	4.28
LGB-T	main	0.083	0.152	3.87	0.121	0.188	4.68
LGB-T	second	0.088	0.151	3.90	0.086	0.166	4.45
LGB-T	third	0.094	0.148	3.89	0.126	0.213	4.51
LGB-C	main	0.086	0.159	4.50	0.120	0.184	4.43
LGB-C	second	0.091	0.160	4.60	0.091	0.173	4.48
LGB-C	third	0.096	0.155	4.48	0.125	0.231	4.65
HGB-Q	main	0.077	0.146	3.35	0.104	0.153	3.23
HGB-Q	second	0.081	0.145	3.38	0.079	0.140	3.19
HGB-Q	third	0.087	0.142	3.29	0.104	0.184	3.20

Method	Window	M5 wQL	M5 median	M5 rank	VN1 wQL	VN1 median	VN1 rank
LGB-Q	main	0.076	0.145	3.10	0.103	0.152	3.18
LGB-Q	second	0.080	0.144	3.17	0.078	0.140	3.13
LGB-Q	third	0.086	0.142	3.15	0.102	0.184	3.14

Nemenyi critical difference (α = 0.05), main / second / third: M5 0.052 / 0.052 / 0.053; VN1 0.077 / 0.084 / 0.093. At h = 13, TSB-NB has the best mean VN1 rank in every window (2.92 / 2.99 / 3.37).

LGB-Q has the lowest wQL at h = 3 in both datasets and all three windows (Table 2); HGB-Q is within 0.002. The same wQL ordering holds within each demand class. On M5, the average per-series rank also favours LGB-Q in every window, by more than the Nemenyi critical difference relative to HGB-Q.

The VN1 rankings are less clear at series level. At h = 3, LGB-Q, HGB-Q and TSB-NB are separated by no more than 0.17 mean ranks, with no significant LGB-Q–HGB-Q difference under the Nemenyi criterion. At h = 13, TSB-NB ranks first in all three windows. Its mean scaled loss on VN1 is 0.336, 0.539 and 2.008 across the windows, inflated by a few series with very small naïve-error denominators; wQL remains 0.103, 0.078 and 0.102.

5.2 Inventory at equal realised fill rate

M5. LGB-Q ranks first on the inventory frontier (Figure 1) in all bootstrap resamples across the three windows. At fill rate 0.94, it needs 3.3–10.6% less stock than LGB-T, LGB-C, TSB-NB and ETS, with the reported intervals excluding zero (Table 3). HGB-Q's inventory requirement differs by 0.2–1.4%.

VN1. The preferred method depends on the window. LGB-Q ranks first in the main window, although its stock advantage over LGB-T, LGB-C and TSB-NB at fill rate 0.94 is not established by the reported confidence intervals. In the second window, TSB-NB ranks first; TSB-NB, LGB-T and LGB-C need 4.1–8.8% less inventory than LGB-Q. In the third, LGB-Q again leads, while TSB-NB, ETS and LGB-T do not attain fill rate 0.94 at any evaluated τ. Lowest wQL does not always mean lowest stock on VN1.

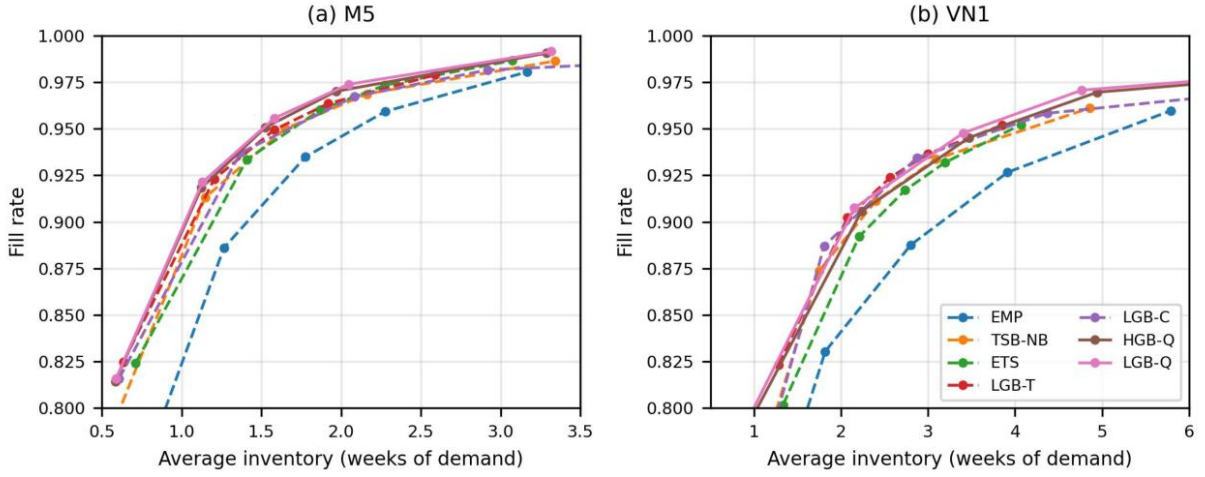


Figure 1. Inventory in weeks of demand against realised fill rate in the main window: (a) M5, (b) VN1. τ ranges from 0.50 to 0.99; solid curves represent direct quantile methods.

Table 3. Inventory needed for fill rate 0.94. Δ = I_method / I_LGB-Q − 1 (%); positive values favour LGB-Q. Brackets show 95% bootstrap intervals; "—" means the method did not reach 0.94.

Set	Window	LGB-Q Inv.	Frontier rank	P(best)	HGB-Q Δ %	LGB-T Δ %	LGB-C Δ %	TSB-NB Δ %	ETS Δ %
M5	main	1.38 [1.34, 1.41]	1.00	1.00	+1.2 [+0.9,+1.6]	+5.1 [+4.4,+5.9]	+4.5 [+3.2,+5.8]	+10.0 [+9.1,+11.0]	+10.6 [+9.8,+11.5]
M5	second	1.49 [1.45, 1.53]	1.20	1.00	+1.4 [+1.2,+1.7]	+6.8 [+6.2,+7.6]	+8.2 [+6.8,+9.3]	+7.5 [+6.7,+8.3]	+8.0 [+7.1,+8.9]
M5	third	1.74 [1.69, 1.78]	1.00	1.00	+0.2 [−0.2,+0.5]	+3.3 [+2.7,+4.0]	+7.5 [+6.3,+8.4]	+4.9 [+4.2,+5.7]	+5.9 [+5.0,+6.6]
VN1	main	3.17 [2.78, 3.57]	1.40	0.93	+4.5 [+2.9,+6.7]	+0.9 [−7.5,+9.9]	+1.7 [−6.3,+8.7]	+10.5 [−0.5,+19.8]	+12.0 [+3.8,+19.7]
VN1	second	1.79 [1.52, 2.05]	3.80	0.00	−1.2 [−2.5,+0.2]	−7.7 [−11.8,−3.8]	−4.1 [−7.3,−0.9]	−8.8 [−12.8,−4.9]	−0.4 [−5.7,+4.2]
VN1	third	3.31 [2.72, 3.85]	1.25	0.95	+1.3 [−0.7,+3.5]	—	+11.5 [+1.6,+116.5]	—	—

LGB-Q requires the least inventory in the smooth, intermittent and lumpy M5 classes in all three windows. On VN1 it leads the lumpy class consistently, but not the smooth class: TSB-NB leads that class twice and LGB-C once.

5.3 Period variation on VN1

Table 4. Demand groups defined retrospectively from test-period demand relative to the preceding 26 weeks (stable: 0.67–1.50×; growing: >1.50×). Δ compares TSB-NB inventory with LGB-Q at fill rate 0.92.

Dataset	Window	Growing share of test demand	Stable Δ %	Growing Δ %	All series Δ %
M5	main	20.7%	+8.5	+12.5	+10.4
M5	second	28.0%	+7.7	+7.7	+5.5
M5	third	19.4%	+3.9	+5.7	+6.1
VN1	main	31.5%	−14.4	TSB-NB not reached	+5.0
VN1	second	22.8%	−13.3	−10.5	−11.3
VN1	third	40.4%	−15.6	TSB-NB not reached	+25.1

For this analysis only, we group series according to the realised ratio of test-window demand to demand in the preceding 26 weeks. Ratios of 0.67–1.50 define the stable group; ratios above 1.50 define the growing group. Because grouping uses outcomes from the test window itself, it cannot be applied as a forecasting-time selection rule.

The stable-demand VN1 group favours TSB-NB in each window: at fill rate 0.92, it uses 13.3–15.6% less inventory than LGB-Q (Table 4). For growing VN1 series, TSB-NB fails to reach that fill rate in the main and third windows, while LGB-Q reaches it. Growing series account for 31.5% and 40.4% of test demand in those windows, compared with 22.8% in the second window, where TSB-NB leads overall. This pattern associates the ranking with demand composition but does not establish causation. On M5, LGB-Q needs less stock in both groups throughout.

5.4 Prospective method selection using demand groups

Table 5. Selection rules trained on one window and evaluated on the next. Numbers are inventory changes (%) relative to always using LGB-Q at fill rate 0.94; negatives favour the rule. Brackets give 95% bootstrap intervals.

Selection rule	M5 third → second	M5 second → main	VN1 third → second	VN1 second → main
Always TSB-NB	+7.5 [+6.7,+8.6]	+10.0 [+8.9,+10.9]	−8.8 [−12.6,−5.1]	+10.5 [+1.9,+20.4]
Best single method in previous window	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	+10.5 [+1.9,+20.4]
By ADI–CV² class	+0.1 [−0.0,+0.2]	+0.0 [+0.0,+0.0]	−5.6 [−8.1,−3.9]	+5.1 [−0.3,+10.2]
By segment	+0.1 [−0.1,+0.3]	+0.0 [+0.0,+0.0]	−0.1 [−0.7,+0.5]	+10.5 [+1.9,+20.4]
By class × segment	+0.1 [−0.0,+0.3]	+0.0 [+0.0,+0.0]	−5.6 [−8.1,−3.8]	+4.8 [−0.1,+12.0]
Oracle by segment (not deployable)	+0.0 [+0.0,+0.0]	+0.0 [+0.0,+0.0]	−8.9 [−12.8,−5.2]	+0.0 [+0.0,+0.0]

We next ask whether information available before a window can identify which forecaster to use. In addition to ADI–CV² classes, we cluster series using eight historical features: level, order size, ADI, CV², time since sale, trend, recent relative level and peak size. ADI, CV² and time since sale are log-transformed; all eight features are then centred on their medians, divided by their interquartile ranges and clipped at ±4, with these statistics taken from the fitting window. The clustering fitted on the earliest window is carried forward unchanged. Predefined criteria select a three-component Gaussian mixture (regular, fading, sparse) that is stable across refits (adjusted Rand index 0.94) but overlapping (silhouette 0.21).

Neither grouping gives reliable improvements (Table 5). On M5, choosing the best method for each group in the previous window changes inventory by no more than 0.3% relative to LGB-Q. For VN1, class-based selection first saves 5.6% and then requires 5.1% more stock; segmentation produces changes of −0.1% and +10.5%. Even a retrospective oracle choosing the best method within each segment cannot beat the best single method of that window. Growing series appear across the groups rather than in a distinct cluster: they represent 18–31% of each VN1 segment and 14–39% of each ADI–CV² class.

5.5 Inventory performance of averaged quantiles

The complementary VN1 results for LGB-Q and TSB-NB suggest testing their average, C2, the arithmetic mean of the two methods' quantiles. We chose these two inputs after inspecting their relative performance, so C2 is exploratory and post hoc. To test combination without that choice, we fixed a second design before running it: the mean (E7) and the median (E7-med) of all seven methods' quantiles, leave-one-out ablations of E7, and a prospective rule that applies the lowest-regret mean subset of one window to the next.

At fill rate 0.92, C2 uses 5.9%, 10.5% and 7.1% less VN1 inventory than LGB-Q in the main, second and third windows and E7-med 7.6%, 8.2% and 7.1% less, with all series-bootstrap intervals excluding zero (but see Section 5.8). On M5 both stay within 1.6% of LGB-Q, whereas E7 needs up to 3.1% more and so misses our preset criterion. Every combination has a higher wQL than LGB-Q. Removing LGB-Q or HGB-Q from E7 raises its stock by 1.0–4.5%, removing ETS lowers it in all six runs, and the lowest-regret subset of every run contains LGB-Q and TSB-NB. At matched realised fill, the prospective rule saves 7.8–8.2% on VN1 in both transitions but needs 0.9% more on M5 in one. C2 also saves VN1 inventory at one- and four-week lead times (Section 5.8).

5.6 Liquidation outcomes

Table 6. Changes relative to no liquidation when LGB-Q supplies the forecasts. Break-even salvage ratios s are calculated under the specified holding-cost and markup assumptions.*

Set	Policy	Window	Liquidated (% demand)	Inventory change (%)	Fill-rate change (pp)	Stockout change (%)	s* range
M5	Quantile	main	0.02	−0.1	−0.00	+0	Unstable (few units)
M5	Quantile	second	0.01	−0.1	−0.00	+0	Unstable (few units)
M5	Quantile	third	0.02	−0.2	−0.00	+0	Unstable (few units)
M5	Fixed	main	0.79	−2.3	−0.29	+7	1.06–1.32
M5	Fixed	second	1.08	−3.0	−0.20	+5	1.01–1.18
M5	Fixed	third	1.21	−2.7	−0.29	+7	1.03–1.23
M5	Dead13	main	0.67	−2.5	−0.65	+13	1.35–2.08
M5	Dead13	second	0.82	−4.9	−0.46	+10	1.21–1.66
M5	Dead13	third	0.90	−3.3	−0.54	+13	1.19–1.70
VN1	Quantile	main	1.18	−4.3	−0.02	+0	0.91–1.04
VN1	Quantile	second	0.27	−1.7	−0.02	+1	0.92–1.21
VN1	Quantile	third	1.74	−7.4	−0.02	+0	0.91–1.04
VN1	Fixed	main	2.76	−6.1	−0.67	+20	1.00–1.22
VN1	Fixed	second	1.47	−5.0	−0.11	+18	0.97–1.07
VN1	Fixed	third	1.91	−4.4	−0.62	+17	1.03–1.29
VN1	Dead13	main	1.21	−4.0	−0.09	+23	1.03–1.37
VN1	Dead13	second	0.76	−3.8	−0.08	+22	1.06–1.32
VN1	Dead13	third	0.47	−2.3	−0.07	+18	1.16–1.60

With LGB-Q forecasts, the quantile rule lowers fill rate by no more than 0.02 percentage points in any window (Table 6). It rarely removes stock on M5; on VN1, it reduces inventory by 1.7–7.4%. Fixed and dead13 liquidation have larger service effects, increasing stockout rates by 5–20% and 10–23%, respectively.

None of the policies has a consistent economic advantage over the 26-week simulation under the assumed costs. For quantile liquidation on VN1, the break-even salvage ratio ranges from 0.91 to 1.21 times unit purchase cost. Ratios for the remaining policies have a lower bound of 0.97. These figures are upper-bound thresholds from the cost model of Section 3.4, not measured resale revenues or realised profits. In practice the rule is direct: liquidate only if the expected recovery ratio r on the secondary market (salvage price divided by unit cost) exceeds s*. With the VN1 quantile-rule thresholds of 0.91–1.21, a channel that recovers 60% of cost would not justify liquidation within 26 weeks, whereas one that recovers about unit cost could. Obsolescence, scarce storage space or a longer horizon would lower s* and are not modelled here.

5.7 Sensitivity to dataset differences

We tested whether specific differences between M5 and VN1 account for the contrasting rankings. Reweighting VN1 to approximate M5's demand-feature distribution reduces the mean standardised difference from 0.32–0.39 to 0.02–0.03 without altering the VN1 findings. Removing VN1's sparse price variables or limiting M5 to its 21 features shared with

VN1 changes LGB-Q inventory by less than 1% in every window. The tested covariate and preprocessing differences do not account for the results; other dataset differences remain untested.

5.8 Robustness checks and a neural baseline

Hyperparameters. Refitting LGB-Q and LGB-T for h = 3 in the main window with small (15 leaves), large (255) and regularised (31 leaves, at least 500 samples per leaf, L2 penalty 10) trees leaves LGB-Q 4.7–6.8% below LGB-T and 7.5–11.0% below TSB-NB in inventory on M5 at fill rate 0.92, with its own inventory changing by less than 3%. The VN1 ranking persists for three settings; with small trees LGB-T uses 8.9% less inventory than LGB-Q.

Lead time. At lead times of one and four weeks in every window (HGB-Q omitted), LGB-Q needs less inventory on M5 than LGB-T, LGB-C, TSB-NB and ETS in all nine window–lead-time runs (2.5–14.8% at fill rate 0.94). On VN1 the post hoc C2 uses 3.2–10.9% less inventory than LGB-Q at fill rate 0.92 in all nine runs, with intervals excluding zero in seven; its maximum regret over the 18 runs is 2.0%, against 12.6% for LGB-Q.

Neural baseline. We evaluated a univariate N-HiTS network [26] separately from the seven-method benchmark. It predicts quantiles of cumulative h-week demand with a multi-quantile loss, is refitted every 13 weeks and uses fixed hyperparameters. It never leads on inventory. At fill rates of 0.90–0.94 on M5 it needs 2.6–10.0% more stock than LGB-Q; its frontier ranks on VN1, among eight single methods, range from 2.8 to 4.4. Including N-HiTS leaves the post hoc C2's maximum regret at 0.9%. It is a CPU-budgeted baseline, not evidence about deep learning as a class.

Computational cost. Except Chronos-2, every method forecasts a 26-week window within minutes on the laptop CPU (TSB-NB 1–2, LGB-Q on M5 10–21, N-HiTS about 7). Zero-shot Chronos-2 [22] needs no training, but its inference took 5.4 hours per M5 window and 32–45 minutes per VN1 window, while needing 14.7% (M5) and 1.2–48.1% (VN1) more inventory than LGB-Q at fill rate 0.92; it ran without fine-tuning or covariates.

Prospective τ, resampling and a third dataset. Choosing each method's τ on the previous window puts every method except EMP within 0.8 points of the target fill on M5, where LGB-T, LGB-C, TSB-NB and ETS then need 3.8–14.2% more inventory than LGB-Q. On VN1 the realised fill misses the target by up to 4.6 points for LGB-Q, HGB-Q, C2 and E7-med; updating τ every week by half the gap between the target and the fill realised so far reduces the miss to 2.4 points, above our one-point criterion. Table 7 separates the target (0.92) from the achieved fill (0.906–0.926 on M5, 0.906–0.940 on VN1, 0.865–0.893 on the footwear panel). Only 13 of 20 comparisons have achieved fills within 0.5 points of LGB-Q's, none on the footwear panel; two of these show a significant saving (E7-med on M5, 0.9%; C2 on VN1, 7.4%), eight need significantly more stock and three are inconclusive, so no method is shown to beat LGB-Q where service differs. Resampling items, stores or blocks of weeks leaves every M5 interval claim intact, whereas warehouse- or client-level resampling leaves only the second-window VN1 savings significant.

Table 7. Past-only service calibration (τ updated weekly, η = 0.5); target fill 0.92 throughout, Fill = achieved fill. ΔI = inventory relative to LGB-Q (%), 95% cluster-bootstrap intervals (M5 items, VN1 warehouses, footwear styles); LGB-Q rows: weeks of demand. † = fills differ by ≥ 0.5 points, not like-for-like.

Method	M5 3→2		M5 2→main		VN1 3→2		VN1 2→main		Footwear	
	Fill	ΔI %	Fill	ΔI %	Fill	ΔI %	Fill	ΔI %	Fill	ΔI %
LGB-Q	0.917	1.21 wk	0.910	1.04 wk	0.908	1.26 wk	0.940	2.95 wk	0.884	1.57 wk
HGB-Q	0.917	+0.4 [+0.2,+0.7]	0.912	+3.5 [+3.1,+3.8]	0.906	−1.2 [−2.3,+0.0]	0.939	+3.9 [+1.2,+6.8]	0.877	−8.8† [−10.7,−6.3]
TSB-NB	0.926	+14.2† [+12.5,+15.9]	0.906	+6.1 [+4.7,+7.6]	0.908	+33.3 [+16.4,+53.7]	0.922	+28.3† [+20.5,+38.2]	0.865	+5.9† [−1.5,+14.5]
C2	0.921	+2.8 [+2.1,+3.6]	0.908	−0.4 [−1.0,+0.2]	0.907	−7.4 [−11.0,−3.8]	0.936	+18.8 [+14.0,+24.4]	0.875	+0.1† [−3.5,+4.6]
E7-med	0.918	+1.1 [+0.5,+1.7]	0.908	−0.9 [−1.6,−0.3]	0.907	−1.1 [−6.2,+5.2]	0.928	+9.6† [+4.5,+15.0]	0.893	+6.0† [+3.8,+8.3]

6 DISCUSSION

The comparison metric changes what counts as a good forecast. With τ fixed at 0.90, the seven methods achieve M5 fill rates ranging from 0.935 to 0.967. Equal nominal quantiles are therefore not equal service outcomes. Matching realised fill rate allows stock requirements to be compared on the same basis. The VN1 window-to-window reversals also show why an inventory ranking based on a single test period may not hold in the next one [7]. Matching realised fill sets τ with the test outcomes, whereas a deployed system must fix it beforehand; on VN1 and the footwear panel the achieved fill then misses the target by up to 5.5 points, so retrospective savings do not become reliable stock reductions, and a ranking should not be transferred without a prospective check.

On VN1, the retrospective demand split helps describe these reversals. TSB-NB uses less stock for stable-demand series, whereas LGB-Q reaches fill rate 0.92 on growing-demand series in two windows where TSB-NB cannot. Theodorou et al. [7] similarly find cases in which forecast accuracy and inventory cost favour different methods, though under different

service and cost settings. If future demand growth were known in advance, assigning LGB-Q to growing series and TSB-NB to the rest would reduce inventory by 5.6–9.8% relative to LGB-Q on VN1. A classifier based on earlier features recovers only part of that hypothetical saving and remains unstable (repository).

Adaptive updating. Parameters are re-estimated only every 13 weeks with fixed hyperparameters, which can leave the global models a block behind level shifts on fast-growing VN1 series. More frequent or warm-started refits, recency weighting or re-tuning could reduce that lag but were not tested.

How the quantiles are constructed also matters: on M5 at fill rate 0.94, LGB-T and LGB-C need 3.3–8.2% more inventory than directly estimated LightGBM quantiles. Residual-based quantiles in [7] and empirical quantiles in [23] may explain part of the disagreement with those studies, which do not isolate this effect.

Among the combinations, E7-med has the smallest maximum regret under the paper's descriptive metric (excess stock over the best single method, averaged over attainable targets): 0.3% across six runs, versus 0.9% for the post hoc C2, 3.4% for E7, 9.4% for LGB-Q, 9.5% for LGB-T, 11.4% for HGB-Q and 20.7% for TSB-NB. When a lost sale costs at most 200 weeks of holding, C2's cost is within 3.1% of the cheapest single model in each run. At higher penalties LGB-Q is cheapest. These figures describe the six evaluated runs, not expected deployment performance.

Toward deployment. Because neither combination passed the prospective checks of Section 5.8, LGB-Q is the safer default. A combination should be frozen and run in shadow mode beside it for several review cycles, with LGB-Q preferred when lost sales cost more than about 200 weeks of holding and upper-quantile coverage monitored, where TSB-NB under-covers.

Censored demand. Both datasets record sales, not demand, and contain no on-hand records. A sales-only detector that flags zero runs between two sales whose probability under the series' sale frequency p is below 1% ($(1 - p)^k < 0.01$) marks 7.4–9.0% of M5 and 6.1–6.9% of VN1 test weeks as suspected stockouts, about twice as often in intermittent and lumpy as in smooth series; because genuine pauses in demand are also flagged, these are upper bounds. Restricting the evaluation to the 72–82% of series without flagged weeks leaves the leading method of every window and C2's lowest maximum regret unchanged. A stricter scenario imputes each series' mean weekly sales in every flagged week. TSB-NB, whose occurrence probability falls during zero runs, then needs 17–27% more inventory than LGB-Q on M5 at fill rate 0.92; the leading method changes in three of six windows, always to a closely ranked method; and C2's maximum regret rises to 9.8%, although it still needs 6.9–10.7% less inventory than LGB-Q on VN1. The pre-specified E7-med is less sensitive: its maximum regret is 0.1% on the clean subset and −2.2% under imputation, where it needs 2.9–5.6% less M5 and 9.5–14.7% less VN1 inventory than LGB-Q at fill rate 0.92. Across detector thresholds of 0.001–0.1 and imputing half the mean, the mean or the mean positive sales, TSB-NB needs 11.8–34.2% more inventory than LGB-Q on M5, while E7-med needs 1.4–7.2% (M5) and 8.2–15.9% (VN1) less and keeps a maximum regret of at most 1.6%. Censoring thus matters most for TSB-NB and for the post hoc case for C2 on brick-and-mortar data; stock-availability records would be needed to settle it. Censoring could also be handled in training, treating flagged weeks as missing or right-censored with an expectation–maximisation step or a censored likelihood [28], censored quantile regression [29] or a survival model of the time between sales; we leave this to future work.

Limitations. Recorded sales stand in for demand, and stockouts are only inferred (see above). We did not tune hyperparameters; the M5 comparison persists across four LightGBM settings, but VN1 is sensitive to one of them (Section 5.8). Growth groups are formed retrospectively. Series-level intervals are too narrow on VN1, where warehouse- and client-level resampling widens them (Section 5.8), and M5 and VN1 supply three evaluation windows, the footwear panel one. Findings may differ for other retailers or inventory policies. Chronos-2 [22] ran zero-shot only. Nabil et al. [27] forecast spare-part demand with a two-stage LightGBM/XGBoost model (an occurrence classifier times a size regressor) and set stock with a normal safety factor; a quantile version of this design is not tested here.

7 CONCLUSION

Under the weekly order-up-to policy tested here, LGB-Q requires the least inventory on M5 for a realised fill rate of 0.94 among the compared statistical and point-based alternatives. Across three test windows, its stock requirement is 3–11% lower, and it also has the lowest volume-weighted quantile loss. This finding persists across the reported lead-time checks and four LightGBM configurations.

The VN1 results do not identify a single leading method across periods. TSB-NB uses less stock on retrospectively identified stable-demand series, while LGB-Q achieves a 0.92 fill rate on growing series in two windows where TSB-NB cannot. Historical demand classes and the tested clustering do not produce consistent benefits when used to select a method prospectively. An equal-weight LGB-Q/TSB-NB quantile average reduces inventory by 6–11% relative to LGB-Q on VN1 at fill rate 0.92 and stays within 1.5% on M5; a pre-specified median of all seven methods saves 7–8% on VN1 with the lowest maximum regret, but the saving does not survive warehouse-level resampling, past-only τ calibration or

an independent footwear dataset, so LGB-Q remains the dependable default. These combination findings are exploratory; the conclusions about single methods above do not depend on them.

Quantile liquidation reduces inventory on VN1 while changing fill rate by at most 0.02 percentage points. Its economic value is not established under the assumed costs over 26 weeks. Useful next tests would use stock-availability records, longer liquidation horizons and more forecasting architectures under the same stock policy.

REFERENCES

[1] S. Makridakis, E. Spiliotis, and V. Assimakopoulos. 2022. The M5 competition: Background, organization, and implementation. International Journal of Forecasting 38, 4, 1325–1336. https://doi.org/10.1016/j.ijforecast.2021.07.007

[2] S. Makridakis, E. Spiliotis, and V. Assimakopoulos. 2022. M5 accuracy competition: Results, findings, and conclusions. International Journal of Forecasting 38, 4, 1346–1364. https://doi.org/10.1016/j.ijforecast.2021.11.013

[3] S. Makridakis, E. Spiliotis, V. Assimakopoulos, Z. Chen, A. Gaba, I. Tsetlin, and R. L. Winkler. 2022. The M5 uncertainty competition: Results, findings and conclusions. International Journal of Forecasting 38, 4, 1365–1385. https://doi.org/10.1016/j.ijforecast.2021.10.009

[4] M. Zanotti. 2025. The cost of ensembling: Is it always worth combining? arXiv:2506.04677.

[5] S. Wang, Y. Kang, E. Spiliotis, and F. Petropoulos. 2026. Multi-objective probabilistic forecast combination for inventory demand. arXiv:2606.04900.

[6] T. E. Goltsos, A. A. Syntetos, C. H. Glock, and G. Ioannou. 2022. Inventory–forecasting: Mind the gap. European Journal of Operational Research 299, 2, 397–419. https://doi.org/10.1016/j.ejor.2021.07.040

[7] E. Theodorou, E. Spiliotis, and V. Assimakopoulos. 2025. Forecast accuracy and inventory performance: Insights on their relationship from the M5 competition data. European Journal of Operational Research 322, 2, 414–426. https://doi.org/10.1016/j.ejor.2024.12.033

[8] Z. Mohammed, C. Anas, and M. El Hammoumi. 2026. A hybrid learning framework for forecasting uncertainty and adaptive inventory planning in retail supply chains. Supply Chain Analytics 13, 100180. https://doi.org/10.1016/j.sca.2025.100180

[9] O. Zabraoui, Y. Hmamou, A. Chafi, and S. Kammouri Alami. 2025. A comparative study of multi-algorithm optimization for inventory analytics in supply chains. Supply Chain Analytics 12, 100154. https://doi.org/10.1016/j.sca.2025.100154

[10] R. H. Teunter, A. A. Syntetos, and M. Z. Babai. 2011. Intermittent demand: Linking forecasting to inventory obsolescence. European Journal of Operational Research 214, 3, 606–615. https://doi.org/10.1016/j.ejor.2011.05.018

[11] A. A. Syntetos, J. E. Boylan, and J. D. Croston. 2005. On the categorization of demand patterns. Journal of the Operational Research Society 56, 5, 495–503. https://doi.org/10.1057/palgrave.jors.2601841

[12] S. Damato, N. Rubattu, D. Azzimonti, and G. Corani. 2026. Intermittent time series forecasting: Local vs global models. arXiv:2601.14031.

[13] N. Kourentzes, J. R. Trapero, and D. K. Barrow. 2020. Optimising forecasting models for inventory planning. International Journal of Production Economics 225, 107597. https://doi.org/10.1016/j.ijpe.2019.107597

[14] J. F. van der Haar, A. P. Wellens, R. N. Boute, and R. J. I. Basten. 2024. Supervised learning for integrated forecasting and inventory control. European Journal of Operational Research 319, 2, 573–586. https://doi.org/10.1016/j.ejor.2024.07.004

[15] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu. 2017. LightGBM: A highly efficient gradient boosting decision tree. In Advances in Neural Information Processing Systems 30.

[16] R. Koenker and G. Bassett. 1978. Regression quantiles. Econometrica 46, 1, 33–50. https://doi.org/10.2307/1913643

[17] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman. 2018. Distribution-free predictive inference for regression. Journal of the American Statistical Association 113, 523, 1094–1111. https://doi.org/10.1080/01621459.2017.1307116

[18] R. Hyndman, A. Koehler, K. Ord, and R. Snyder. 2008. Forecasting with Exponential Smoothing. Springer. https://doi.org/10.1007/978-3-540-71918-2

[19] F. Pedregosa et al. 2011. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830.

[20] J. Demšar. 2006. Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research 7, 1–30.

[21] N. Vandeput. 2024. VN1 Forecasting – Accuracy Challenge. DataSource.ai.

[22] A. F. Ansari et al. 2025. Chronos-2: From univariate to universal forecasting. arXiv:2510.15821.

[23] E. Spiliotis, S. Makridakis, A. Kaltsounis, and V. Assimakopoulos. 2021. Product sales probabilistic forecasting: An empirical evaluation using the M5 competition data. International Journal of Production Economics 240, 108237. https://doi.org/10.1016/j.ijpe.2021.108237

[24] F. Petropoulos, X. Wang, and S. M. Disney. 2019. The inventory performance of forecasting methods: Evidence from the M3 competition data. International Journal of Forecasting 35, 251–265. https://doi.org/10.1016/j.ijforecast.2018.01.004

[25] M. Z. Babai, M. M. Ali, and K. Nikolopoulos. 2012. Impact of temporal aggregation on stock control performance of intermittent demand estimators: Empirical analysis. Omega 40, 713–721. https://doi.org/10.1016/j.omega.2011.09.004

[26] C. Challu, K. G. Olivares, B. N. Oreshkin, F. Garza, M. Mergenthaler-Canseco, and A. Dubrawski. 2023. N-HiTS: Neural hierarchical interpolation for time series forecasting. In Proceedings of the AAAI Conference on Artificial Intelligence 37, 6. https://doi.org/10.1609/aaai.v37i6.25854

[27] O. M. Nabil, N. H. Afia, and T. Ismail. 2026. Intermittent demand forecasting for spare parts using classification/regression models of machine learning and deep learning. Ain Shams Engineering Journal 17, 104445. https://doi.org/10.1016/j.asej.2026.104445

[28] S. Nahmias. 1994. Demand estimation in lost sales inventory systems. Naval Research Logistics 41, 6, 739–757. https://doi.org/10.1002/1520-6750(199410)41:6<739::AID-NAV3220410605>3.0.CO;2-A

[29] J. L. Powell. 1986. Censored regression quantiles. Journal of Econometrics 32, 1, 143–155. https://doi.org/10.1016/0304-4076(86)90016-3